RUDRA PRATAP SINGH

AI/ML and Full-Stack Developer | Generative AI, AI Agents, RAG, LLM Evaluation and AI Security

Bengaluru, Karnataka | rudra19867@gmail.com | linkedin.com/in/rudra-pratap-singh-5a0ab532a | github.com/rudraksh01-tech |
Portfolio: rudraksh01-tech.github.io

SUMMARY

Computer Science undergraduate (VTU, CGPA 8.0) building Generative AI and Large Language Model (LLM) applications: AI agents, RAG, LLM evaluation and AI security. Shipped three live, tested projects (Python, LangGraph, FastAPI, React, PostgreSQL, Docker, Kubernetes); Founding Software Engineer Intern at FERMOR; seeking AI and Machine Learning (AI/ML) internships.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering (CSE) 2024 - 2028 (Expected) | 5th semester
Cambridge Institute of Technology, North Campus, Bengaluru | VTU Current CGPA: 8.0 / 10

EXPERIENCE

Founding Software Engineer Intern, FERMOR | *Bengaluru, Karnataka (Hybrid)* Sep 2026 - Oct 2026

- Redesigned key website pages end to end with a cleaner frosted-glass UI/UX, and optimised page structure, headings, metadata and content for Search Engine Optimization (SEO).
- Drove about 30x the usual daily traffic on launch day (a month's worth in one day) through the redesign and SEO work; used web analytics to measure the impact and guide next steps.

PROJECTS

Dwarpal | AI Security Gateway for LLM Agents (MCP) rudraksh01-tech.github.io/dwarpal
Python, LangGraph, MCP, FastAPI, PostgreSQL, pgvector, Redis, Docker, Kubernetes, Helm | github.com/rudraksh01-tech/dwarpal

- Built an open-source AI security gateway (guardrails) between LLM agents and MCP tool servers that blocks prompt injection, tool poisoning, rug pulls and sensitive-data leaks in English, Hindi, Hinglish and Kannada.
- Designed a LangGraph policy graph with human-in-the-loop approval: SHA-256 tool pinning, cross-server data-flow (taint) tracking, multilingual rules, the Llama Prompt Guard 2 classifier and an LLM-as-a-judge, with Redis caching.
- Evaluated over 60 repeated agent runs: the undefended agent leaked user data in 5 of 25 attack runs, 0 of 25 with Dwarpal and 0 false blocks; showed Prompt Guard 2 scored a politely worded attack 0.001 (safe).
- Stored tool pins and audit logs in PostgreSQL with pgvector; shipped Docker images and a Helm chart that CI/CD (GitHub Actions) deploys and tests on Kubernetes; 34 unit and integration tests.

AyurIP-Sahayak | Multilingual RAG Assistant ayur-ip-sahayak.vercel.app
Python, FastAPI, React, Google Gemini API, NumPy, Docker, GitHub Actions | github.com/rudraksh01-tech/AyurIP-Sahayak

- Built a Retrieval-Augmented Generation (RAG) assistant for Ayurveda and Indian IP law that answers in English, Hindi, Kannada and Hinglish with page-level citations and streaming responses.
- Engineered retrieval from scratch (PDF ingestion, section-aware chunking, Gemini embeddings, vector search, BM25 + Reciprocal Rank Fusion hybrid search): Hit@1 0.95 and MRR 0.97 on 38 labelled questions vs 0.47 and 0.60 for BM25.
- Built an LLM-as-a-judge evaluation: 0.99 faithfulness (166 of 167 claims), 0.98 citation precision, 15 of 15 planted errors caught; fixed out-of-memory deploys by replacing a local PyTorch model with an embeddings API; 47 tests.

NestFinder | Agentic AI Assistant for Flat Search nestfinder-woad.vercel.app
Node.js, Express.js, React, MongoDB, Groq API, Langfuse, OpenTelemetry | github.com/rudraksh01-tech/nestfinder

- Developed a tool-calling (function calling) AI agent from scratch: 5 tools, streaming, structured JSON outputs, a repair loop for invalid tool calls and automatic model fallback on API rate limits.
- Wrote a 20-case agent evaluation covering Hindi, Hinglish and prompt-injection inputs; fixed a schema bug it exposed, raising tool-choice accuracy from 18/20 to 20/20.
- Added LLM observability with OpenTelemetry and Langfuse (PII masked), JWT authentication, rate limiting and MongoDB geospatial queries; deployed on Vercel and Render with CI.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL | **Frontend:** React, Vite, Tailwind CSS, HTML, CSS

Generative AI and LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Agentic AI, LangGraph, Model Context Protocol (MCP), Tool Calling (Function Calling), Prompt Engineering, Structured Outputs (JSON Schema), Embeddings, Natural Language Processing (NLP), Google Gemini API, Groq API

Retrieval and Vector Search: Vector Databases (pgvector), Vector Search, Semantic Search, Hybrid Search (BM25, RRF), Reciprocal Rank Fusion, NumPy

LLM Evaluation and AI Security: LLM Evaluation, LLM-as-a-Judge, Guardrails, Prompt Injection Defense, OWASP Top 10 for LLMs, LLM Observability

Backend and Databases: FastAPI, Node.js, Express.js, REST APIs, Async Python, Pydantic, PostgreSQL, Redis, MongoDB, JWT

DevOps and Cloud: Docker, Kubernetes, Helm, CI/CD (GitHub Actions), Git, Linux, Vercel, Render, Unit Testing (pytest)

Core CS: Data Structures and Algorithms, Object-Oriented Programming (OOP), Operating Systems, DBMS, Computer Networks